A-A-50224
September 22, 1986

COMMERCIAL ITEM DESCRIPTION

EXPOSURE CONTROL, PHOTOLITHOGRAPHIC, LIGHT-INTEGRATING

The General Services Administration has authorized the use of this commercial item description in preference to Military Specificaion MIL-E-13500.

This description covers a light-integrating-type exposure control unit, use to automatically control the quantity of light required to expose lithographic plates or photographic negatives regardless of fluctuations in the light source.

Salient characteristics:

The light-integrating photolithographic exposure control (hereinafter called "exposure control") shall consist of a main control unit, a photoelectric cell unit, remote control unit, and all necessary connectors and cables required for proper operation. The exposure control shall be a multi-range instrument with a basic range and one or more secondary ranges. When the dial is set at the maximum range position on the basic range scale and the photoelectric cell window is illuminated with 1,000 foot-candles, an exposure time of 200 seconds, plus or minus 10 percent, shall result. The secondary range or ranges shall provide an exposure duration of at least 10 minutes. If filters are used, they should not bleach nor darken significantly through aging or exposure to light within the useful operating life of the equipment.

The exposure control shall be stable within a time range from 10 seconds to 10 minutes and a voltage range from 95 to 125 volts, with a light intensity of 1,000 foot-candles on the photoelectric cell. The exposure control shall operate on 95-125-volt, 50-60-Hertz (Hz), single phase current and shall contain solid state or all-transistorized circuitry. The instrument shall be protected with a properly rated replaceable-type fuse or circuit breaker. Five spare fuses shall be furnished with each instrument. All components of the exposure control shall be grounded, and comply with the National Electric Code.

Unless otherwise specified, the exposure control shall be the GAM III exposure computer (with accessories including multicolor phototube, Ortho phototube, GAM Flash-Fax, remote control station, and color phototube control switch) or equal.

AMSC N/A FSC 6760

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:41Z
Check the source to verify that this is the current version before use.

Main control unit. The main control unit shall be wall-mountable and shall be equipped with an "on-off" power switch or button, pilot light to indicate when power is "on", a switch or button to initiate the exposure a means of instantly cancelling an exposure, and an elapsed-time indicating dial to manually set various exposure durations. Provisions shall be made on the instrument to make repeated identical exposures. The power cable shall not be less than 10-feet (3.048 meters (m)) long and shall be equipped with a three-prong, grounded, male plug. The main control unit shall have no dimension longer than 7-inches (0.178m). All control switches and electrical plugs shall be labeled to full describe function. The unit shall be labeled to fully describe voltage.

Photoelectric cell unit. The photoelectric cell shall be a solid-state or photodiode type. The case shall have a window of light-diffusing glasss through which all light must pass to reach the cell. Two spare light-diffusing glasses shall be furnished. The cell shall be housed in a dustproof metal case having a cubage of not more than 6 cubic inches (.0001 m^3) and with the shorter dimension on the window side of the case being not nore than 1.5 inches (0.038 m). A cable not less than 25-feet (7.620 m) long and equipped with a suitable quick-connect type plug for connection to the main control unit shall be provided. The photoelectric cell case shall be provided with a mounting bracket so the photoelectric cell may be removed without tools. The mounting bracket shall be provided with straps with drilled countersunk holes or keyhole slots positioned to permit permanent mounting of the unit on the front of the printing frame or camera.

Remote control unit. The remote control unit shall be equipped with a switch to initiate the exposure and an auxiliary "on-off" switch to activate the shutter. All controls shall be labeled to fully describe their function. A power cable at least 10-feet (3.048 m) long, with a quick-connect type plug, shall be provided for connection to the main control unit. The unit shall be housed in a dustproof case having a cubage not more than 43 cubic inches (.0007 m^3), and no dimension longer than 7 inches (0.178 m). Metal straps with drilled countersunk holes or keyhole slots shall be provided on the outside of the case to permit mounting of the unit on a wall.

Regulatory requirements. The offeror/contractor in encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Packaging and marking. Military packaging shall be specified when shipments are to overseas Department of Defense (DoD) consignees. Commercial packaging may be specified for shipments to DoD consignees within the continental United States and to non-DoD consignees worldwide. When a supplier's normal packaging will satisfy the requirements for military packaging, it may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:41Z
Check the source to verify that this is the current version before use.

Military packaging. Each exposure control unit shall be preserved and packed to protect against physical damage and deterioration in a logistic environment characterized by the following:

1. Multiple handling during transportation and intransit storage.
2. Shock, vibration, and static loading forces during worldwide shipment by all transport modes.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and intransit transfers.
5. Stacking loads during shipment and storage.

Quantities of exposure control units destined to a single addressee may be unitized into one or more unit loads.

Commercial. Each exposure control unit shall be packaged in accordance with the supplier's normal method for retail distribution. Exposure control units shipped to a single destination may be unitized to permit handling as a single unit. Shipping containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

Marking. Unit, intermediate (when provided), and exterior packs shall be marked with the item nomenclature, stock or part number, quantity per pack, and contract number as a minimum; additional marking shall be as specified in the contract or order. Exterior packs shall have in addition the complete consignee's address. Marking shall be applied in a legible manner in a contrasting color, of an appropriate size, and of waterproof materials.

Quality assurance. Each exposure control shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Ranges and exposure time not as specified.
2. Spare fuses not supplied as specified.
3. Grounding and compliance with National Electric Code not as specified.
4. Main control unit not as specified.
5. Photoelectric cell unit not as specified.
6. Connecting cables not as specified.
7. Remote control unit not as specified.
8. Packaging and marking not as specified.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial market place for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:41Z
Check the source to verify that this is the current version before use.

Small business. A small business that cannot meet the certification-of-commercial-item requirement of this document, may make an offer under this solicitation; and, if such small business is the apparent successful offeror, the Government may require the small business to submit a sample photolithographic exposure control. Any such exposure control may be evaluated for this solicitation if time to contract award permits. Such evaluation would be to determine conformance to all requirements of this commercial item description and to determine equivalency to photolithographic exposure controls available in the commercial marketplace in the areas of durability, human factors engineering, and safety. It should be noted that bid samples are not required unless specifically requested by the contracting officer. Note. The Government may elect not to evaluate a bid sample from an apparent low bidder for award for this solicitation if a determination of urgency is made by the contracting officer. The bid sample would then be evaluated prior to future solicitation for comparable items.

Notes.

1. A GAM III or equal exposure control will satisfy the requirements of this CID. The GAM III is available from the Graphics Arts Manufacturing Company, 2518 South Boulevard, Houston, Texas 77098.
2. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.
3. The ordering activity shall specify exposure control components and accessories required. The procuring activity shall supply the ordering activity with available commercial literature to facilitate inclusion of necessary accessories in ordering documents.

MILITARY INTEREST:

Custodians
Army - ME
Air Force - 99

Review activity
Air Force - 70
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FCGC

PREPARING ACTIVITY
Army - ME

Project 6760-0576

Source: https://assist.dla.mil -- Downloaded: 202610-26T08:41Z
Check the source to verify that this is the current version before use.